

Numerical Simulation for Evolution of Water Meniscus in Unsaturated Clay using Peridynamics

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① Background

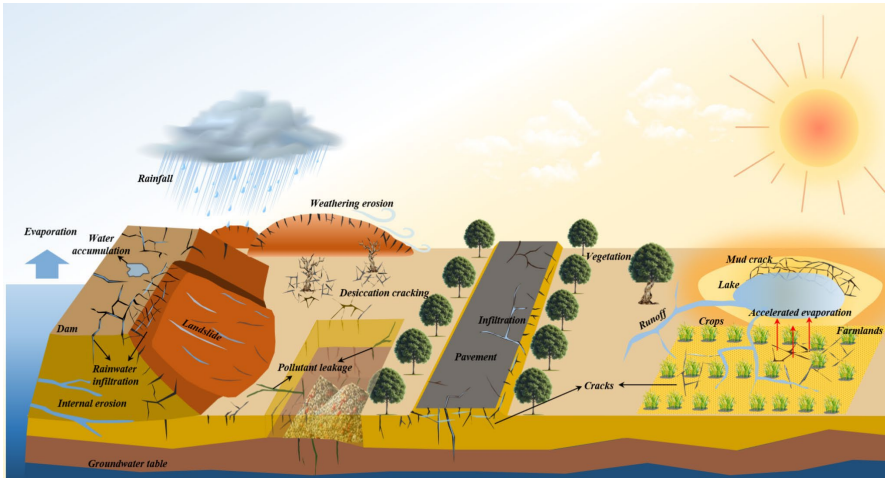
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④ Summary

Motivation

Desiccation cracking of clay is intensified by global warming and extreme drought.



Motivation

Desiccation cracking of clay is intensified by global warming and extreme drought.

Cracking of different soil bodies in drought disasters

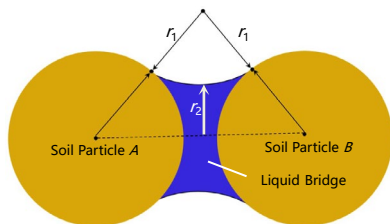


(a) Cracking of clayey soil in Texas, USA in 2022



(b) Cracking of sandy soil in the Kingdom of Morocco, Africa in October 2024

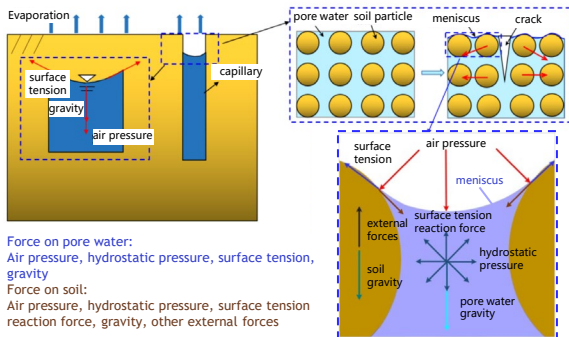
The single liquid bridge model of unsaturated clay



The matrix suction of unsaturated soil is defined by Young-Laplace Equation, related to the morphology curved liquid surface

$$s = \gamma_s \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

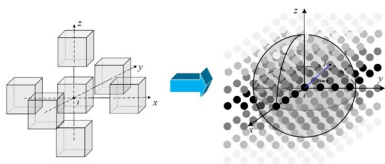
Micro-scale meniscus evolution controls capillary suction → governs macro-scale shrinkage and cracking!



Limitations of existing numerical methods

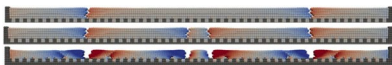
Research Methods	Model	Features	Limitation
DDA	General Model	The soil mass is discretized into irregular polygonal particles	The solid-liquid contact angle and radius of curvature of a meniscus are constant, making it impossible to obtain information about its dynamic evolution
	Two-dimensional extend model	Determining the morphology of the meniscus based on the water content per unit volume of soil	
DEM	General model	The effect of surface tension is replaced by adhesive force	Lack of description of menisci
	Isolated capillary model	Suitable for two-dimensional soils with low saturation	Introducing numerous parameters to describe the morphology of the meniscus may introduce significant calculation errors and fails to capture the dynamic evolution process
	Cableway liquid bridge model	Suitable for highly saturated two-dimensional soils	
	Complex fluid cluster model	Suitable for highly saturated three-dimensional soils	
CFD-DEM	One-way fluid-structure interaction model	Using CFD method to simulate water pressure, using DEM method to simulate soil particle movement, considering unsaturated soil matrix suction using constant adhesion force	The flow field is not affected by the movement of soil particles and cannot fully reflect the fluid-structure interaction under surface tension
MD	General model	Nanoscale liquid bridge model	Modeling and calculation based on molecular force fields Lack of description of surface tension

PD modeling in soil deformation

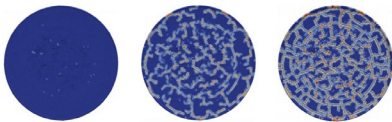


Local model of
continuum
mechanics

Non-local model
of peridynamics
(Silling, 2000)

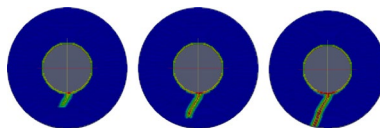


Evolution of cracking in soil strip

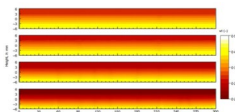


Evolution of crazing in soil plate

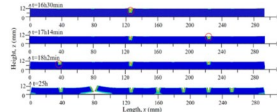
(Menon and Song, 2019)



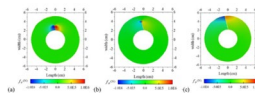
(Jabakhani, 2013)



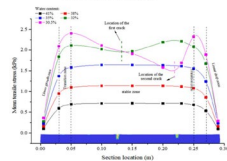
Moisture transport in soil strip



Cracking of soil strip



Evolution of stress in soil ring cracking



Distribution of stress in soil strip cracking

(Yan et al., 2023; Liu et al., 2023)

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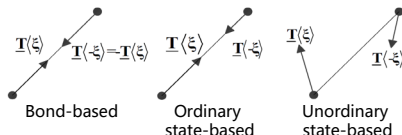
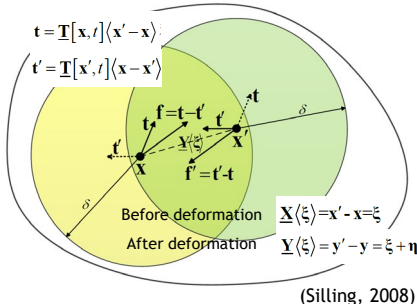
② **Methods**

③ Results & Discussion

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PD framework and PDDO

Interaction between material points in PD modeling framework



Peridynamic differential operators

$$\frac{\partial^{n_1+n_2+\dots+n_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \dots \partial x_M^{p_M}} = \int_{H_x} f(\mathbf{x} + \xi) g_N^{p_1 p_2 \dots p_N}(\xi) dV_{\mathbf{x}'}$$

(Madenci et al., 2016)

Governing equations
of free surface flow

→
PDDO

Non-local form
expression

Lagrangian description

PD integral equations

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= -\rho \nabla \cdot \mathbf{v} \\ \rho \frac{\partial \mathbf{v}}{\partial t} &= -\nabla P + \mathbf{F}_V + \mathbf{F}_B + \mathbf{F}_S \end{aligned}$$

where

$$P = \frac{\rho_0 c_f^2}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right) + P_0$$

$$\mathbf{F}_V = \mu(\Delta \mathbf{v})$$

$$\mathbf{F}_B = 0$$

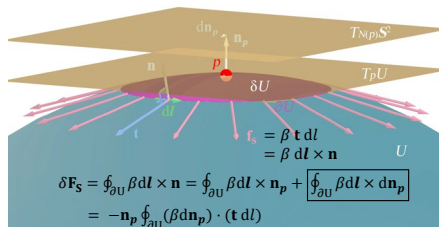
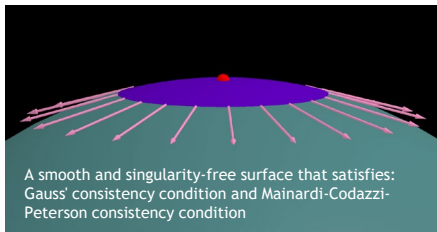
$$\mathbf{F}_S = \beta \kappa \nabla s$$

$$\begin{aligned} \frac{\partial \rho(\mathbf{x})}{\partial t} &= -\rho(\mathbf{x}) \int_{H_x} \mathbf{g}_1(\xi) \cdot (\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})) dV_{\mathbf{x}'} \\ \rho \frac{\partial \mathbf{v}}{\partial t} &= - \int_{H_x} (P(\mathbf{x}') - P(\mathbf{x})) \mathbf{g}_1(\xi) dV_{\mathbf{x}'} \\ &\quad + \int_{H_x} \mu(\mathbf{x}) (\mathbf{v}(\mathbf{x}')) \\ &\quad - \mathbf{v}(\mathbf{x}) \text{tr}(\mathbf{g}_2(\xi)) dV_{\mathbf{x}'} \\ &\quad - \beta \int_{H_x} \kappa (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{\mathbf{x}'} \end{aligned}$$

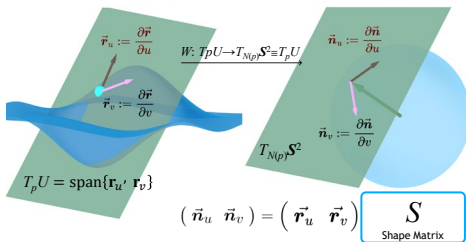
where

$$\kappa = \frac{\int_{H_x} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\xi) dV_{\mathbf{x}'}}{\int_{H_x} w_0(\|\xi\|) dV_{\mathbf{x}'} / \int_{H_x} w_0(\|\xi\|) dV_{\mathbf{x}'}}$$

Calculation of surface tension

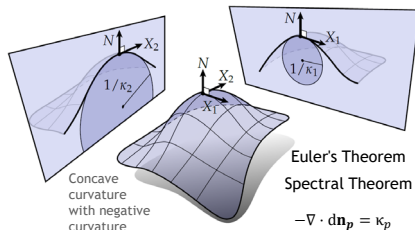


$$\vec{\mathbf{r}}(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \\ z(u, v) \end{pmatrix} \xrightarrow{N: U \rightarrow S^2 \subset \mathbb{R}^3} \frac{\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v}{\|\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v\|} \quad \boxed{\vec{\mathbf{n}}(u, v)} = \begin{pmatrix} n_1(u, v) \\ n_2(u, v) \\ n_3(u, v) \end{pmatrix}$$



$$\text{Gauss theorem} \Rightarrow \mathbf{n}_p \oint_{\partial U} (\beta d\mathbf{n}_p) \cdot (\mathbf{t} dl) = \beta (\nabla \cdot d\mathbf{n}_p) \delta U \mathbf{n}_p$$

$$\mathbf{F}_S = \lim_{\delta U \rightarrow 0} \frac{\delta \mathbf{F}_S}{\delta U} = \beta (-\nabla \cdot d\mathbf{n}_p) \mathbf{n}_p$$



Euler's Theorem
Spectral Theorem

$$-\nabla \cdot d\mathbf{n}_p = \kappa_p$$

Calculation process

Continuity Equation

$$\rho_i^{n+1} = \rho_i^n - \rho_i^n \Delta t \sum_{j=1}^{N_i} [(\mathbf{v}_j^n - \mathbf{v}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n)] V_j$$

$$\rho_i^{n+1} = \rho_i^n - \frac{2\Delta x^2}{\pi\delta^2} \rho_i^n \Delta t \sum_{j=1}^{N_i} (\mathbf{v}_j^n - \mathbf{v}_i^n) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|}$$

For two-dimensional problems, the relative value form of the near-field dynamic differential operator is adopted.

Momentum equation

$$\mathbf{a}_i^{n+1} = \frac{1}{\rho_i^{n+1}} \left(-\sum_{j=1}^{N_i} (P_j^n - P_i^n) \mathbf{g}_1(\xi_{ij}^n) V_j + \sum_{j=1}^{N_i} \mu_i (\mathbf{v}_j^n - \mathbf{v}_i^n) \text{tr}(\mathbf{g}_2(\xi_{ij}^n)) V_j \right) + \frac{1}{\rho_i^{n+1} \mathbf{g} \mathbf{B}}$$

$$- \beta \frac{\left(\sum_{j=1}^{N_i} \min(N_i, N_j) (\mathbf{n}_j^n - \mathbf{n}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n) V_j \right)}{\sum_{j=1}^{N_i} \min(N_i, N_j) w_0(\|\xi_{ij}^n\|) V_j / \sum_{j=1}^{N_i} w_0(\|\xi_{ij}^n\|) V_j} \left(\sum_{j=1}^{N_i} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi_{ij}^n) V_j \right) + \frac{1}{\rho_i^{n+1} \mathbf{g} \mathbf{B}}$$

$$- \sum_{j=1}^{N_i} (P_j^n - P_i^n) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|} + \frac{3\mu_i}{\delta} \sum_{j=1}^{N_i} \frac{(\mathbf{v}_j^n - \mathbf{v}_i^n)}{\sqrt{\|\xi_{ij1}^n + \xi_{ij2}^n\|}}$$

$$\mathbf{a}_i^{n+1} = \frac{2\Delta x^2}{\pi\delta^2 \rho_i^{n+1}} \left(-\frac{\left(\sum_{j=1}^{N_i} \min(N_i, N_j) (\mathbf{n}_j^n - \mathbf{n}_i^n) \cdot \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|} \right) \left(\sum_{j=1}^{N_i} (s(\mathbf{x}') - s(\mathbf{x})) \frac{[\xi_{ij1}^n \ \xi_{ij2}^n]^T}{\|\xi_{ij1}^n + \xi_{ij2}^n\|} \right)}{\frac{\pi\delta^2}{2\beta\Delta x^2} \left(\sum_{j=1}^{N_i} \min(N_i, N_j) w_0(\|\xi_{ij}^n\|) V_j / \sum_{j=1}^{N_i} w_0(\|\xi_{ij}^n\|) V_j \right)} \right) + \frac{1}{\rho_i^{n+1} \mathbf{g} \mathbf{B}}$$

Time difference scheme

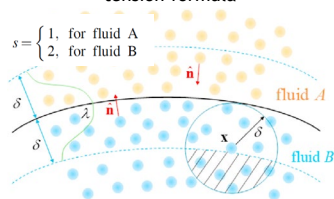
Second-order Verlet-Velocity algorithm

$$\begin{cases} \mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \frac{1}{2} (\mathbf{a}_i^{n+1} + \mathbf{a}_i^n) \Delta t \\ \mathbf{u}_i^{n+1} = \mathbf{u}_i^n + \mathbf{v}_i^n \Delta t + \frac{1}{2} \mathbf{a}_i^n \Delta t^2 \\ \mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{u}_i^{n+1} \end{cases}$$

Courant-Friedrichs-Levy stability condition

$$\Delta t < \min \left(\frac{h_{\text{smooth}}}{|\mathbf{v}_{\text{max}}| + c_B}, \frac{1}{8} \frac{h_{\text{smooth}}^2}{\max(\mu/\rho_0)}, \frac{1}{4} \sqrt{\frac{h_{\text{smooth}}}{g}}, \frac{1}{4} \sqrt{\frac{\min(\rho_0) h^3}{2\pi\beta}} \right)$$

The non-local form of the surface tension formula



$$(s(\mathbf{x}') - s(\mathbf{x})) = \begin{cases} 2\rho_{x,0}, & \text{if } \mathbf{x} \text{ and } \mathbf{x}' \text{ belong to different phases} \\ \rho_{x,0} + \rho_{x',0}, & \text{if } \mathbf{x} \text{ and } \mathbf{x}' \text{ belong to the same phase} \end{cases}$$

Introduce the truncation function to avoid the unit normal vector field having incorrect directions

$$\mathbf{n}(\mathbf{x}) = \begin{cases} \frac{\int_{H_K} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{K'}}{\int_{H_K} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{K'}}, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$$

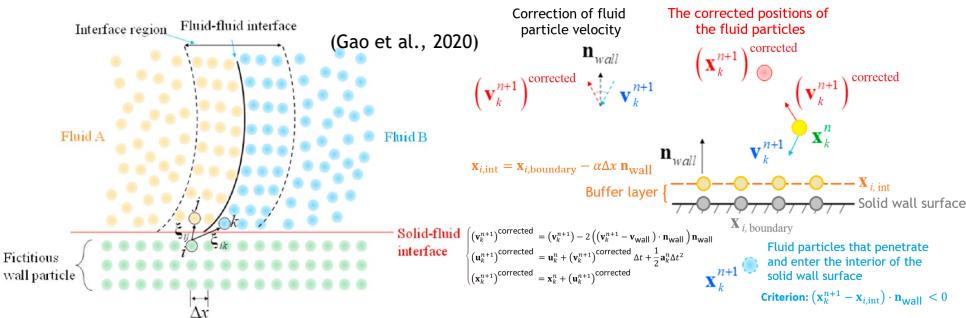
$$\kappa^*(\mathbf{x}) = -\nabla \cdot \mathbf{n}(\mathbf{x}) = - \int_{H_K} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\xi) dV_{K'}$$

Introduce correction factors to correct the curvature field of boundary particles let $\mathbf{N}(\mathbf{x}) = \begin{cases} 1, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$

$$\zeta^*(\mathbf{x}) = \frac{\int_{H_K} \min(\mathbf{N}(\mathbf{x}), \mathbf{N}(\mathbf{x}')) w_0(\|\xi\|) dV_{K'}}{\int_{H_K} w_0(\|\xi\|) dV_{K'}}, \kappa^{**}(\mathbf{x}) = \frac{\kappa^*(\mathbf{x})}{\zeta^*(\mathbf{x})}$$

(Gao et al., 2020)

Boundary handling



Considering the no-slip boundary, the fluid velocity is zero at the wall surface.

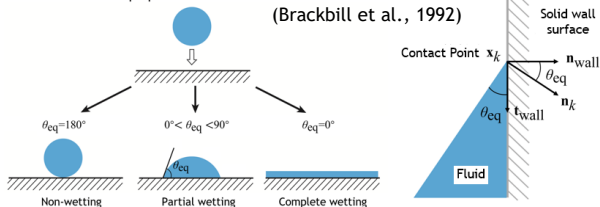
$$\mathbf{v}_i^n = 2\mathbf{v}_{wall} - \sum_{k=1}^{N_f} w_0(\|\xi_{ij}^n\|) \mathbf{v}_k^n$$

$$P_i = \frac{\sum_{k=1}^{N_f} (P_k + (\mathbf{F}_k^B - \rho_k \mathbf{a}_{wall}) \cdot \xi_{ik}^n) w_0(\|\xi_{ij}^n\|)}{\sum_{k=1}^{N_f} w_0(\|\xi_{ij}^n\|)}$$

$$\rho_i = \rho_{0,i} \left(\frac{P_i - P_{0,i}}{P_{ref,i}} + 1 \right)^{1/\gamma_i}$$

Consider the surface properties of the solid

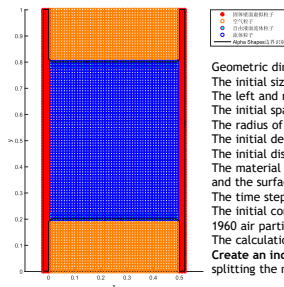
(Brackbill et al., 1992)



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Numerical example



Geometric dimensions: $0.5 \text{ cm} \times 1.0 \text{ cm}$.

The initial size of the central liquid bridge is $0.5 \text{ cm} \times 0.6 \text{ cm}$.

The left and right boundaries are stationary rigid solid walls.

The initial spacing between the material points $\Delta x = 0.01 \text{ cm}$.

The radius of the near-field domain $\delta = 3.015 \Delta x$.

The initial density of the fluid particles is set to $\rho_0 = 1 \text{ kg/m}^3$.

The initial displacement, velocity, acceleration and pressure are all zero.

The material constant of the fluid $\gamma = 1$, the artificial sound speed $c_f = 12 \text{ cm/s}$, the dynamic viscosity coefficient $\mu = 0.2 \text{ Ps}\cdot\text{s}$, and the surface tension coefficient $\beta = 1 \text{ N/m}$.

The time step $\Delta t = 5.0 \times 10^{-7} \text{ s}$, and the total number of time steps $N = 15000$.

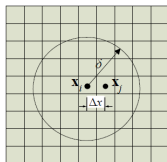
The initial computational domain was discretized into 2989 fluid particles, 606 fictitious particles representing solid walls, and 1960 air particles.

The calculation program for the example was written using **MATLAB 2015b**.

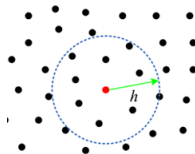
Create an independent main program file that encompasses all the calculation processes, rather than adopting the method of splitting the main program into multiple sub-program scripts and then calling them.



<https://github.com/Liskelleo/Numerical-Simulation-of-the-Evolution-of-Water-Meniscus-Based-on-Peridynamics>

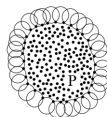


Two-dimensional structure with central orthogonal uniform discretization.

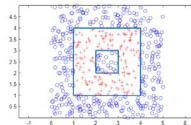


By calling the library function **rangesearch** to conduct a KD-tree neighborhood search for the specified particle.

The free surface is identified by calling library functions such as:



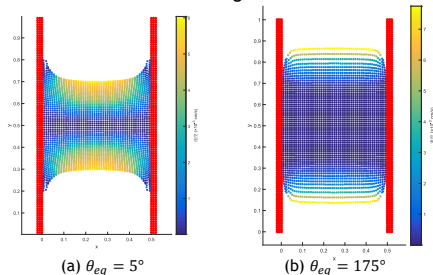
alphaShape



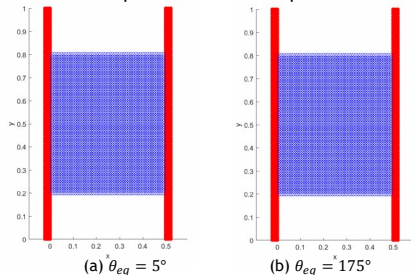
inpolygon

Results

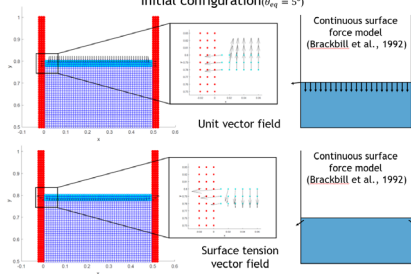
Final configuration



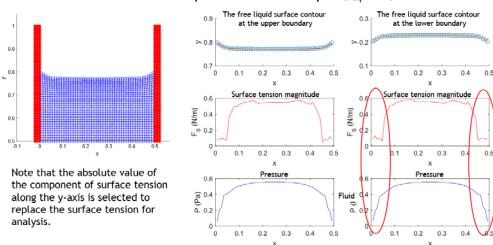
Liquid level evolution process



Initial configuration ($\theta_{eq} = 5^\circ$)

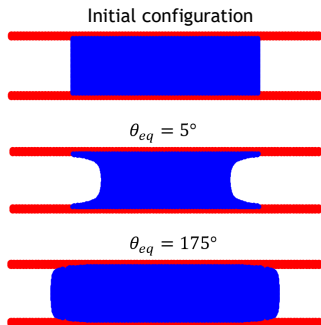


Particle distribution at 7500 time steps for the concave liquid surface example ($\theta_{eq} = 5^\circ$)

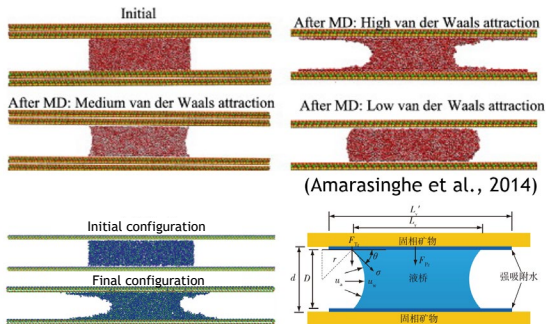


PD vs. MD

The stronger the adsorption of the solid boundary on the pore water, the more "concave" the curved liquid surface will be; conversely, it will be more "convex".



(a) PD results



(Luo et al., 2022)

(b) MD results

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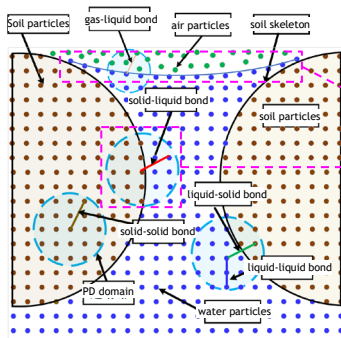
Future Work

Simplified single liquid bridge model

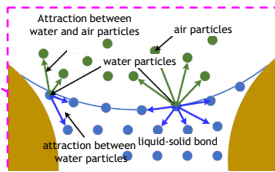
Capillary action ✓

Evaporation effect ✗

Investigation on the Evolution of Curved Liquid Surface under the Coupling Effect of Heat, Water and Force is our future work!



The microscopic structure of clay

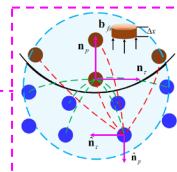


One-way gas-liquid interaction

Boundary conditions:

Gas $\rightarrow$ Liquid: Surface tension

Solid $\rightleftharpoons$ Liquid: Decomposes into two forces acting in the pressure direction and the shear direction



Fluid-solid interaction

Thanks for your time and attention!